

Influence of process parameters on microstructure and interfacial mechanical properties of Al6061/AlSi10Mg multi-material components fabricated via laser powder bed fusion

Linda Ismail ^a, Omer F. Mohamed ^b, Tareq Farrah ^c, Pradeep George ^b,
Andreas Schiffer ^{a,b,*}

^a Department of Mechanical and Nuclear Engineering, Khalifa University of Science and Technology, Abu Dhabi, 127788, United Arab Emirates

^b Advanced Research and Innovation Center (ARIC), Khalifa University of Science and Technology, Abu Dhabi, 127788, United Arab Emirates

^c Research Laboratories, Khalifa University of Science and Technology, Abu Dhabi, 127788, United Arab Emirates

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ABSTRACT

Advanced manufacturing has evolved in recent years through the synergistic integration of additive manufacturing (AM) with conventional methods to enable the creation of high-performance components with complex features. In this study, AlSi10Mg powder was printed on extruded Al6061-T6 substrate blocks using Laser Powder Bed Fusion (LPBF). An orthogonal L16 Taguchi experiment was designed where the laser power, scanning speed, and hatching distance of the first ten layers of the interface were varied to examine their isolated effect on the bonding quality of multi-material structures. This interfacial bond quality was evaluated through Iosipescu shear testing and various characterization techniques. Laser power was found to be the principal factor influencing defect morphology: lack of fusion (LOF) voids, keyhole pores, and crack formation; followed by Linear Energy Density (LED). A maximum average interfacial shear strength of 178.54 ± 5.49 MPa was achieved by the bimetallic parts, close to that of AlSi10Mg. Moreover, an inverse correlation was observed between shear strength and the porosity percentage in the fusion region. The AlSi10Mg region exhibited the highest micro-hardness (109.56 ± 0.5 HV), while Al6061 had the lowest (98.20 ± 1.26 HV), and the interface showed intermediate values (99.35 ± 1.03 HV).

1. Introduction

Aluminum alloys, specifically AlSi10Mg, have garnered significant attention in the AM field. Fully dense parts of these alloys have been successfully produced using LPBF, focusing on optimizing their mechanical properties through adjustments in process parameters. For example, various studies have thoroughly investigated how printing parameters affect the mechanical properties of LPBF AlSi10Mg to enhance its performance [1,2]. Fine-tuning of scanning speed and laser power was found to improve the density, hardness, elongation, and strength of the formed parts [3,4]. Moreover, post-processing, such as heat treatment, could influence mechanical and electrochemical properties. For instance, a group of researchers studied the impact of annealing heat treatment on the corrosion resistance of LPBF AlSi10Mg, noting that annealing at 300 °C resulted in abnormally low corrosion

resistance [5].

Multi-material AM structures were introduced to achieve tailored mechanical, chemical, and physical characteristics in specific areas of a single component by merging the distinct properties of various materials [6]. However, traditional joining methods such as welding, diffusion bonding, and powder alloying encounter difficulties with the fabrication of multi-material structures due to high cost, low efficiency, and limitations in attainable geometries [7]. In contrast, AM offers significant design flexibility and rapid production capabilities, making it an attractive option for creating multi-material parts and innovative designs. This capability primarily benefits the automotive, biomedical, and aerospace industries, which demand complex, high-performance components [8]. Among the AM techniques, LPBF stands out for its ability to produce intricate metallic multi-material structures. It achieves this through very thin layers and focused laser spots, allowing for the precise

* Corresponding author. Department of Mechanical and Nuclear Engineering, Khalifa University of Science and Technology, Abu Dhabi, 127788, United Arab Emirates.

E-mail address: andreas.schiffer@ku.ac.ae (A. Schiffer).

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fabrication of complex designs for end-use components [9].

Extensive research has been conducted on fabricating aluminum AM multi-material parts using LPBF, which has broad applications in repair and mold manufacturing. For instance, a study exploring the mechanical properties of Al-Cu-Ni-Fe-Mg/AlSi10Mg bimetallic parts demonstrated that the LPBF exhibited impressive mechanical performance with higher microhardness and improved tensile properties [10]. In a different study by Fathi et al. [11], the corrosion behavior of an AA2618/AlSi10Mg multi-material LPBF part was examined in 3.5 wt.% NaCl solution, where the LPBF segment exhibited superior corrosion resistance compared to the AA2618/AlSi10Mg multi-material and AA2618 cast samples. In another research work, Zhao et al. [12] investigated the influence of heat treatment on the mechanical and corrosion properties in 3.5 wt.% NaCl solution of AA6061/AlSi10Mg multi-material. After the T6 heat treatment, the multi-material part showed a bonded interface with a minor effect on the texture. Despite a decrease in microhardness at the interface, it exhibited moderately high corrosion resistance.

Ensuring reliable interfacial bonding between dissimilar materials in multi-material structures is challenging due to their varying thermo-physical properties, such as thermal conductivity, laser absorptivity, and melting points [13], as well as chemical incompatibility. The mismatch in these properties can cause defects at the interface, including delamination, unmelted powder particles, and cracks [14]. These defects can weaken the structure and lead to premature failure of multi-material components during service. For instance, Bahduri et al. [15] showed that as-built AA6082/AlSi10Mg bimetal parts failed at the interface despite achieving the ultimate tensile strength of fully aged AA6082. Post-heat treatment analysis revealed notable microstructural enhancements, including reduced porosity, improved joint strength, and increased microhardness, likely due to the formation of Mg₂Si precipitates. Another study investigated the effect of the ratio of the LPBF part volume to the substrate volume on the tensile properties of the AA6061/AlSi10Mg multi-material components [16]. Results showed that the strain localized within the interface region in samples with 50 % or more LPBF part. In contrast, samples with less than 50 % LPBF part exhibited higher uniformity in the strain field across both the interface and the substrate. Thus, a bimetallic structure's mechanical reliability and performance mainly depend on the characteristics of the interfacial joint, particularly its bond strength. In addition to the tensile strength, interfacial shear strength is another aspect of evaluating the bonding strength between different materials. Moreover, accurate shear testing helps determine the ability of these interfaces to resist shear forces, providing insights into their mechanical performance and potential failure modes. Few studies have investigated the interface bonding of multi-material parts by conducting shear tests [17,18]. However, research on assessing the interfacial bonding of aluminum multi-material LPBF parts through shear testing remains scarce, as noted in ZainElabdeen et al. [19]. Given that the substrate, interface, and LPBF part are in an isostress state under tensile loading, the failure of the sample is likely to occur in the weakest segment, which may not be at the interface. This aligns with previous studies [10,12,15,16] which report failure in the weaker base material rather than at the interface in similar hybrid components. Consequently, relying only on tensile testing may not provide a precise evaluation of the bonding strength unless the interface is indeed the weakest link. This study aims to overcome these challenges by utilizing shear testing, which offers a more direct and comprehensive understanding of the bonding in bimetallic parts. This is particularly crucial for ensuring the structural integrity and reliability of multi-material components used in demanding applications.

In this study, AlSi10Mg powder is printed on pre-formed Al6061-T6 substrates using the LPBF process, with the aim of fabricating a fully dense multi-material part with a strong bonding at the interface. The effect of process parameters, such as laser power, scanning speed, and hatching distance, on the bond strength is examined through an L16 orthogonal experiment design. The microstructure and morphology of

the fusion region are characterized, and the mechanical properties of the multi-material parts are further investigated through Iosipescu shear testing and microhardness measurements. This work reports the first application of the Iosipescu shear test in multi-material AM, presenting an efficient technique that allows for an accurate assessment of interfacial mechanical properties under shear loads and addresses a significant gap in the literature. Additionally, this work includes a thorough interfacial parametric study for the Al6061/AlSi10Mg combination, offering new insights into the influence of processing parameters on interfacial strength and failure mechanisms.

2. Experimental

2.1. Materials and LPBF process

AlSi10Mg powder was printed on eight extruded Al6061-T6 substrate blocks, each having dimensions of 40 mm × 40 mm × 55 mm, using an EOS M400-4 metal 3D printer (EOS GmbH, Germany). CNC face milling was used to ensure the surface flatness of the substrate blocks to ±30 μm, which is necessary to ensure the smooth motion of the re-coater and an even layer thickness in all samples. For this purpose, a face milling head equipped with carbide milling inserts (Garant SOMET09T304TR, HB735) was used at a cutting speed of 7000 rpm, a feed rate of 2000 mm/min (0.0571 mm/tooth), and a cutting depth of 1 mm. A coolant was used during the face milling process to avoid overheating of the surface and minimize the formation of an oxide layer.

Gas-atomized AlSi10Mg powder was procured from Powder Alloy Corporation (PAC, USA). Fig. 1 shows the morphology and the particle size distribution of the AlSi10Mg powder, which was measured using a Phenom XL G2 scanning electron microscope (SEM) image with a sample size of 1895 particles. Particle size ranged between 4.9 and 77.6 μm, with a standard deviation of 8.96 μm. The manufacturer does not supply the optimum parameters for mono-material PAC AlSi10Mg. Therefore, they were determined by the authors in an unpublished full factorial study comprising 75 samples. The optimum build parameters were chosen based on the part density, calculated using micro-CT imaging, and tensile properties. A maximum density of 99.993 %, with a corresponding ultimate tensile strength (UTS) of 409.3 MPa and an elongation of 4.19 %, was achieved for the optimum parameter specimen. These optimum parameters were used for the bulk of the additively manufactured part and are outlined in Table 1.

In this work, a 4-level, 3-factor Taguchi design of experiment (DoE) was used to study the individual effects of laser power, scanning speed, and hatching distance on the fusion zone density and the shear strength of the bimetallic bond. The process parameters used for the first ten layers are shown in the L16 orthogonal array of Table 2. The first two layers in each sample were remelted once each to ensure complete melting. The layer thickness was fixed to 0.03 mm throughout the build process.

Process parameters control the energy delivered to the powder bed, influencing the melt pool dimensions and dynamics, which ultimately govern defect formation and the structural integrity of the bond between the two parts. Therefore, the Linear Energy Density (LED) and Volumetric Energy Density (VED) were calculated for each set of process parameters as follows:

$$LED = \frac{P}{v} \quad (1)$$

$$VED = \frac{P}{v h t} \quad (2)$$

where P is the laser power (W), v is scanning speed (mm/s), h is the hatching distance (mm), and t is the layer thickness (mm).

It is important to note that the blocks were placed in an aluminum fixture and secured with screws throughout the printing process. To achieve consistent results, care was taken to ensure that the build

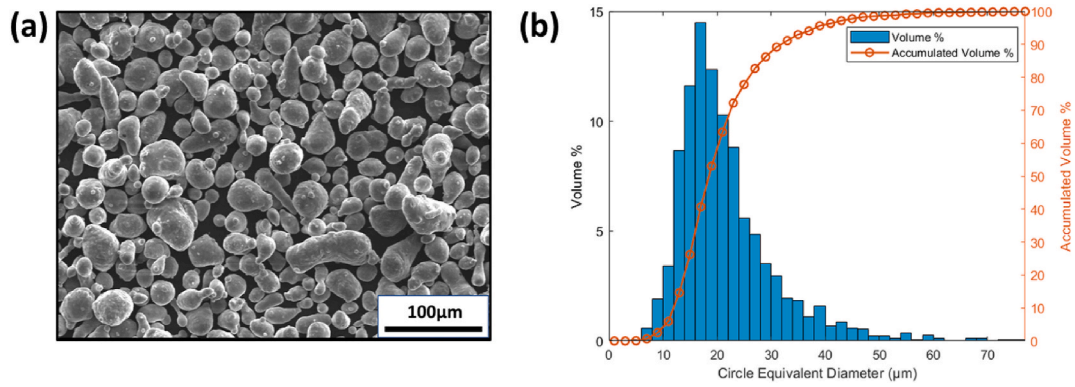


Fig. 1. (a) SEM image showing the morphology of AlSi10Mg powder. (b) Particle size distribution of the AlSi10Mg powder.

Table 1

Printing Parameters used for AlSi10Mg after the first ten layers.

	Power [W]	Scanning Speed [mm/s]	Hatching Distance [mm]	Layer Thickness [mm]
AlSi10Mg	275	1500	0.09	0.03

Table 2

L16 orthogonal array showing the process parameters used for the first 10 interfacial layers and the associated LED and VED.

Sample no.	Power [W]	Scanning Speed [mm/s]	Hatching Distance [mm]	Linear Energy Density [J/mm]	Volumetric Energy Density [J/mm ³]
1	250	1000	0.06	0.250	138.9
2	250	1250	0.09	0.200	74.1
3	250	1500	0.12	0.167	46.3
4	250	1750	0.15	0.143	31.7
5	275	1000	0.09	0.275	101.9
6	275	1250	0.06	0.220	122.2
7	275	1500	0.15	0.183	40.7
8	275	1750	0.12	0.157	43.7
9	300	1000	0.12	0.300	83.3
10	300	1250	0.15	0.240	53.3
11	300	1500	0.06	0.200	111.1
12	300	1750	0.09	0.171	63.5
13	325	1000	0.15	0.325	72.2
14	325	1250	0.12	0.260	72.2
15	325	1500	0.09	0.217	80.2
16	325	1750	0.06	0.186	103.2

surface of each block was aligned with the print bed at the same height to ensure uniform layer thickness across all parts. The build plate and substrate blocks were pre-heated to 165 °C for 2 h before initiating the printing process. A bi-directional scanning strategy with a rotation of 67° between layers was used throughout the printing process. The as-built bimetallic specimens are depicted in Fig. 2(a).

2.2. Microstructural characterization

A block for microstructural characterization was cut out from each specimen using wire electrical discharge machining (EDM), according to the schematic in Fig. 2(b). To examine the interfacial morphology of the multi-material samples, each characterization block was ground and polished in three stages in accordance with the “Metallographic Preparation of Aluminum Alloys” notes from Struers. A mixture of 100 ml water, 6 ml nitric acid, and 3 ml hydrofluoric acid was used as an etching solution.

Subsequently, the interface features were observed using an Optika optical microscope (OM), a Phenom XL G2 SEM, and a JEOL JSM-7610F

FEG-SEM. The electron dispersive spectroscopy (EDS) in the Phenom SEM was used to determine the chemical composition and elemental distribution along the fusion zone. The grain size and morphology in the interfacial region were characterized using the electron backscatter diffraction (EBSD) detector in a Quanta 3D SEM. The surface roughness of the substrate after face milling was evaluated using Bruker Alicona InfiniteFocus, an optical 3D surface measuring system, over a surface area of 6 mm² and a profile length of 4 mm.

High-resolution imaging of multi-material Al6061/AlSi10Mg samples was performed using a Phoenix Nanotom Micro-CT system (GE, USA). The region of interest (ROI) for porosity measurement was positioned at the interface, such that it encompassed 0.5 mm depth from each material, as illustrated in Fig. 3. This ROI volume and position were maintained across all samples. The images were analyzed with VGStudio Max software to calculate the porosity and assess interfacial integrity.

2.3. Iosipescu shear test

The strength of the interfacial bond in shear loading was evaluated using a Iosipescu shear test fixture in Fig. 2(c), supplied by Wyoming Test Fixtures in accordance with the ASTM D5379 standard [21]. While the fixture is commonly used for composites due to the limited range available for deformation, it has been successfully used to evaluate the shear strength of Al6061 in several studies [22,23]. The shear specimen dimensions used are illustrated in Fig. 2(d). The shear test was conducted using an MTS Alliance RF/150 machine with a load cell of 150 kN capacity and a cross-head speed of 2 mm/s. Each sample was tested three times to ensure that the results were consistent. Digital image correlation (DIC) GOM software was used with a standalone camera setup to evaluate the strain field within the interface region. The Iosipescu shear test samples were sprayed white and then speckled with black spray paint to allow for deformation tracking in the DIC software. Shear stress was calculated using:

$$\tau_i = P_i / A \quad (3)$$

where P_i is the load at the i th data point, and A is the cross-sectional area of the minimum area region. The strain was measured by drawing two 1.5 mm ‘digital extensometers’ placed at the center of the specimen at angles of $\pm 45^\circ$ to the x -axis. Then, the shear strain was measured by adding the two strain measurements:

$$\gamma_i = |\epsilon_{+45}| + |\epsilon_{-45}| \quad (4)$$

where ϵ_{+45} and ϵ_{-45} are the $+45^\circ$ and -45° normal strain at i th data point, respectively.

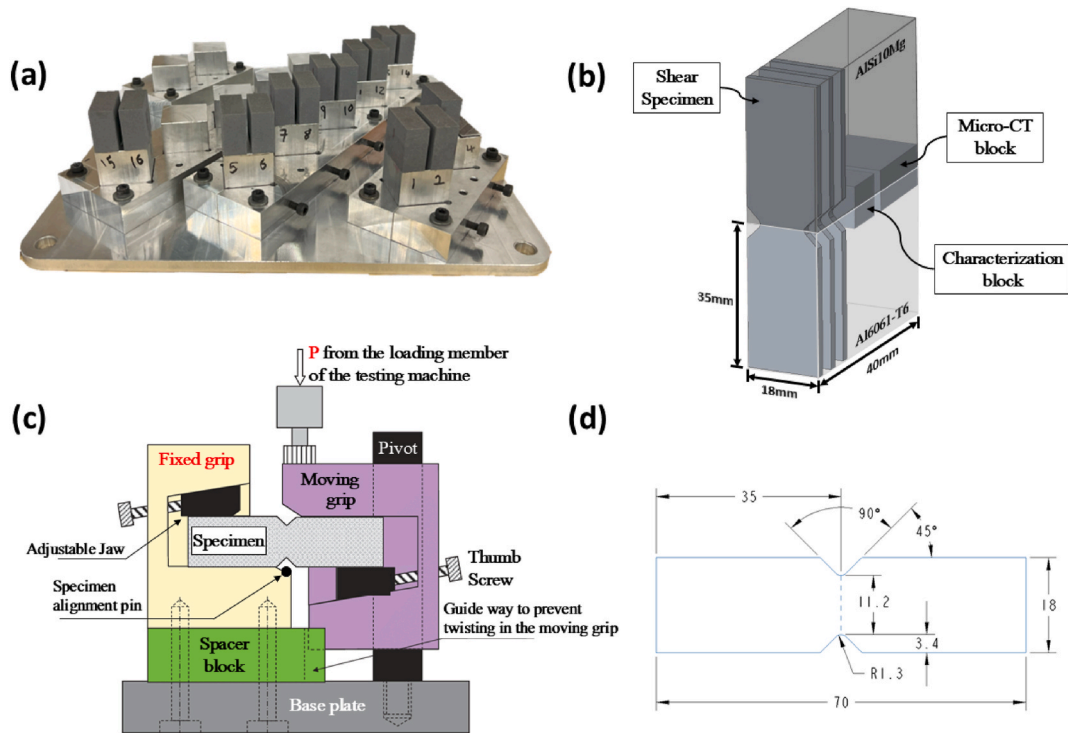


Fig. 2. (a) Post-built multi-material parts in fixture setup (b) Location of shear specimens and blocks used for microstructural characterization and micro-CT imaging (c) Iosipescu Shear Test Setup [20] (d) Iosipescu shear test specimen dimensions.

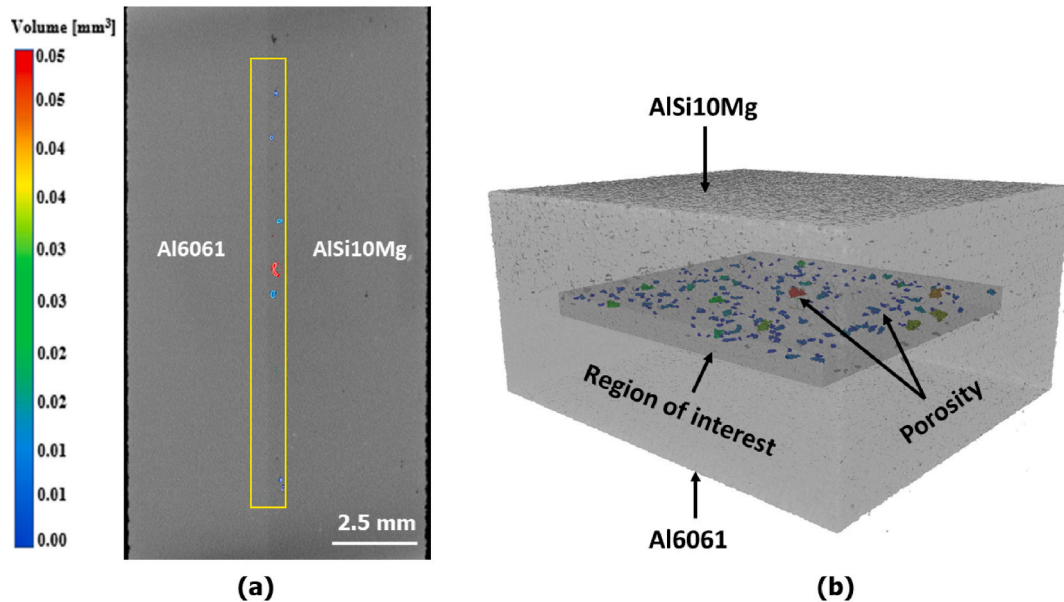


Fig. 3. Micro-CT image showing the region of interest (yellow rectangular) used in the calculation of the porosity of the multi-material part (a) 2D cross-sectional view and (b) 3D view. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3. Results and discussion

3.1. Microstructure and interfacial characteristics

Fig. 4 depicts the defect morphology and density at the interface of Al6061/AlSi10Mg bimetallic parts built with various LPBF process parameters, highlighting trends with changing laser power and linear energy density. A clear transition from lack of fusion (LOF) defects at the top left corner to hot cracks and keyhole pores at the bottom right corner is evident. Significant LOF defects and delamination at the interface can

be observed in samples 3 and 4, respectively. This observation may be attributed to the low LED and VED associated with those samples. The delamination of sample 4 resulted in the protrusion of the additive section and subsequent collision with the re-coater, which required the premature termination of the build. Similarly, the weak interfacial bonding in sample 3 resulted in premature fracture during wire EDM cutting of the shear test samples. Therefore, the shear properties of these two samples could not be evaluated.

As can be observed from Fig. 4, at a laser power of 250W, increasing LED and VED from samples 4 to 2 led to a significant decrease in the size

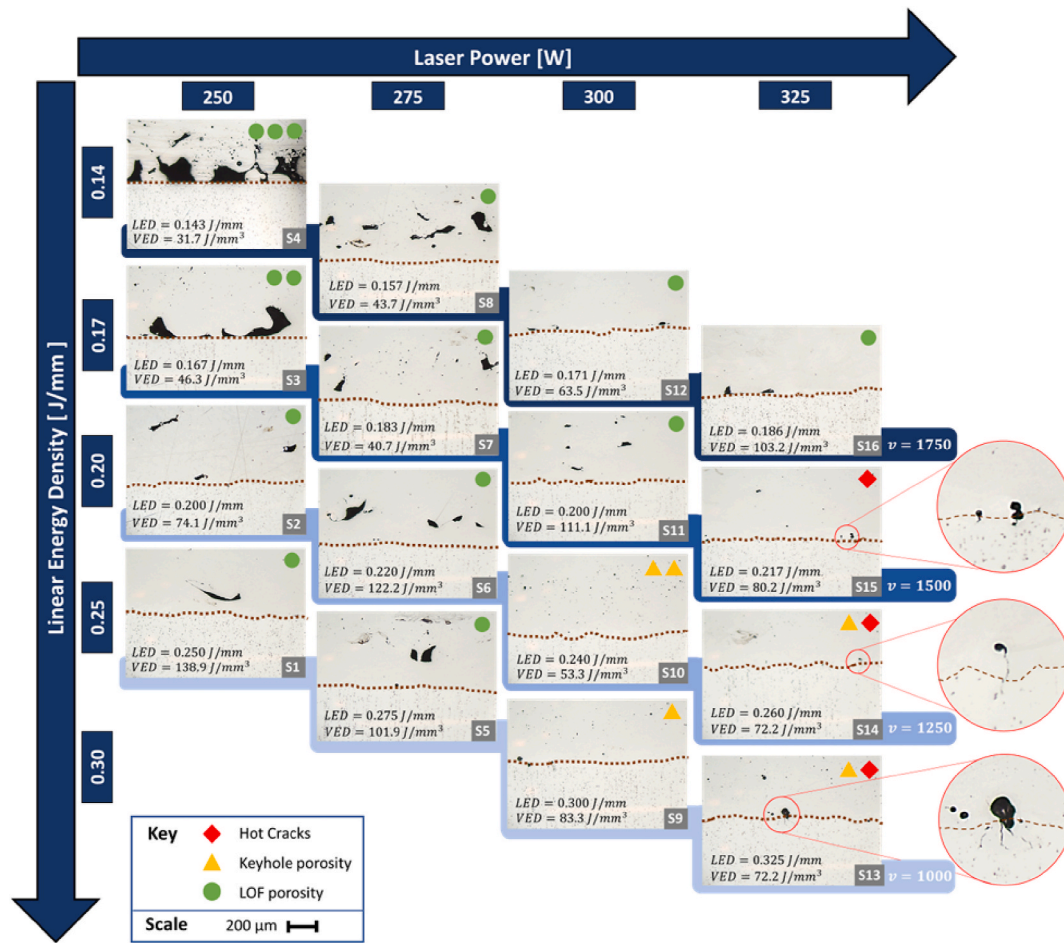


Fig. 4. The effect of power and linear energy density on the density and morphology of defects in the L16 samples. The interface is identified through a brown dashed line, and the number of symbols at the top-right corner signifies the intensity of the specific defect in each sample. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

and density of LOF voids. Sample 2 shows incomplete melting with visible unmelted powder particles, suggesting insufficient energy density. Contrary to expectation, increasing both LED and VED from sample 2 to sample 1 did not result in the elimination of LOF porosity; where instead, several horizontal crack-like voids with a width of 100–300 μm were formed in the vicinity of the interface. As reported in Ref. [24], these defects often form due to insufficient bonding between subsequent layers caused by the shallow melt pools, which cannot remelt a portion of the previously consolidated layer. This effect can be observed in samples 1, 11, 12 and 16.

A similar effect is observed in samples 4, 8, 12, and 16, which all have a scanning speed of 1750 mm/s, where increasing the LED and VED from S4 to S16 resulted in a noticeable decrease in the size and density of LOF voids. However, there was no significant improvement when increasing the laser power from 300 W to 325 W. It is further observed that samples 12 and 16 are nearly defect-free in the fusion zone, except for a few horizontal LOF defects at the first layer between the additive part and the substrate.

It was not possible to eliminate LOF voids at $P = 250$ W and $P = 275$ W despite the relatively high LED and VED used in some samples, such as S1, S5, and S6. Meanwhile, at higher laser powers of 300 W and 325 W, a clear transition from LOF voids to keyhole pores was observed when decreasing the scanning speed from 1500 mm/s to 1250 mm/s. This highlights the significance of power over both LED and VED in determining the defect morphology.

A high density of small-sized keyhole pores was exclusively present in samples 9, 10, 13, and 14, which were built at the highest power,

300 W and 325 W, and the lowest scanning speeds, 1000 mm/s, and 1250 mm/s, considered in this study. The formation of keyhole pores seems to be more dependent on the laser power rather than LED or scanning speed because the higher LED associated with samples 1 and 5 failed to generate any keyhole porosity. Most notably, the increase of keyhole porosity with a decrease of VED in this set of parameters indicates the insufficiency of VED as an indicator of keyhole porosity.

A similar trend was observed at a laser power of 300 W (see Fig. 4), where decreasing the scanning speed initiated a transition from LOF voids, at scanning speeds of 1750 mm/s and 1500 mm/s, to keyhole porosity, at scanning speeds of 1250 mm/s and 1000 mm/s. As discussed earlier, the low LED in samples 11 and 12 resulted in horizontal crack-like pores.

A limited presence of <50 μm in length microcracks was observed exclusively at the highest laser power of 325 W at variable scanning speeds of 1000 mm/s, 1250 mm/s, and 1500 mm/s (samples 13–15). Considering that other samples with higher VED and LED but lower power did not experience any cracking, such as samples 1 and 5, it appears that cracking is also primarily influenced by the laser power and then, to a lower extent, by the LED or scanning speed of the track.

Similar cracking behavior was reported by Wang et al. [16] during the fabrication of an Al6061/AlSi10Mg multi-material component when using a substrate remelting step. The large mushy zone of 6000 series Al Alloys makes them naturally susceptible to solidification cracking. Therefore, the melting of the Al6061 substrate must be kept at a minimum. Moreover, the authors suggested that the high Si content of AlSi10Mg would decrease the width of the mushy zone, eliminating

crack formation, which was achieved when no remelting was used.

From Fig. 4, it is evident that cracking occurred exclusively at the highest power, 325W, regardless of the volumetric energy density of the process parameters used. However, LED had a secondary role in determining the likelihood of crack formation, where cracks occurred at and above an LED of 0.217 J/mm. Similarly, the horizontal crack-like voids in sample 1 occurred despite the relatively high energy densities used in this sample. In comparison, samples built at lower LED and VED than sample 1, such as samples 10 and 15, showed a contrasting defect morphology, with keyhole pores and hot cracking, as opposed to the LOF defects in sample 1. Similarly, samples 5 and 6, which were built at relatively high LED and VED, showed the formation of large LOF pores; in contrast to samples 14 and 15, which were built at similar LEDs to samples 5–6, but displayed a completely different defect type. Moreover, at a constant scanning speed of 1000 mm/s, an unexpected transition from conduction to keyhole melting is observed with decreasing VED. From the above observations, it is clear that LED and VED cannot be independent indicators of the melting mode or defect type, and that laser power plays a pivotal role in determining both the melting mode and defect type. This observation agrees with the findings of Gan et al. [25], who defined a dimensionless keyhole number to determine the stability of the melt pool dynamics as a function of material properties and the process parameters (laser power P , scan speed V_s , and laser radius r_0). The equation defined by Gan et al. highlights the superior effect of laser power relative to scan speed on the stability and dynamics of the melt pool.

The preferential etching of the hybrid samples revealed the melt pool morphology, the extent of intermixing at the interface, and the precise positioning of defects relative to both the melt pool and the interface, as illustrated in Fig. 5. Circulate flows, also known as Marangoni convection, are clearly visible in the vicinity of the interface. Marangoni convection, which is responsible for the mixing of materials within the melt pool, is a function of the surface tension gradient ($d\gamma/dT$) and the temperature gradient within the melt pool ΔT . The higher energy density applied to sample 13 has led to a more homogenous mixing at the interface, in contrast to the lower energy density in sample 3, which led to a sharper interface with macro-segregation between the two miscible materials. Conversely, the increased mixing between Al6061 and AlSi10Mg at the fusion region in samples 13 and 14 has led to the formation of hot cracks.

The substrate surface condition has a significant effect on the bonding strength in hybrid parts. Both Fan et al. [26] and Kuai et al. [27]

investigated the effects of substrate surface treatments on the interface characteristics of hybrid parts. Kuai et al. [27] observed that the most pristine interface was achieved at the lowest surface roughness of the 316L substrate ($S_a = 0.23 \pm 0.07 \mu\text{m}$). Meanwhile, Fan et al. [26] discussed the importance of surface wettability in addition to surface roughness, particularly in Aluminum alloy substrates, where the elevated temperatures involved in some surface preparation methods, such as EDM, may result in the formation of Al_2O_3 oxides. Despite achieving the lowest roughness ($R_a = 2.7 \pm 0.4 \mu\text{m}$) using EDM, these oxides decreased the surface wettability and absorptivity of laser power, decreasing the melting pool temperature at the interface, and resulting in unmelted particles and lack-of-fusion pores [28]. In this study, surface milling was used to ensure the flatness of the substrate and minimize its surface roughness. Surface roughness measurements of $S_a = 0.581 \mu\text{m}$ and $R_a = 0.297 \mu\text{m}$ were achieved, with a maximum peak height of $8.972 \mu\text{m}$ and a minimum peak height of $-2.928 \mu\text{m}$. These surface roughness values correlate with the findings of Singh et al. [29], $R_a = 0.257\text{--}0.363 \mu\text{m}$, for various ‘near-dry’ Minimum Quantity Lubrication (MQL) parameters using constant cutting speed of 5000 RPM and feed-rate of 0.2 mm/tooth. The authors determined a strong correlation between cutting temperature and surface roughness, where a higher cutting temperature increased surface roughness, likely due to the increased formation of oxides. Upon comparison with the milling parameters, surface roughness, and temperature measurements by Singh et al. [29], and considering the use of a flooding coolant in this study, rather than the near-dry conditions in MQL, it is deduced that the face milling was performed at sufficiently low temperatures to prevent the formation of a thick oxide layer.

The EDS analysis of sample 14, illustrated in Fig. 6(a), demonstrates a homogenous distribution of elements across the interfacial region, indicating strong bonding with minimal elemental segregation. The distribution of silicon at the interface reveals the mixing pattern between the Al6061 and AlSi10Mg through circulate flows caused by Marangoni convection. The multi-material sample exhibited the least porosity and fewest defects and showed a consistent presence of key alloying elements throughout the interface, such as aluminum, silicon, and magnesium. The uniform element distribution could enhance interfacial integrity and mechanical performance [15]. However, it can be noticed that there is element segregation on the Al6061 side, particularly silicon and magnesium. This could be attributed to the extrusion process, where inhomogeneous deformation in the extrudate may lead to a non-uniform distribution of Mg-Si particles, resulting in

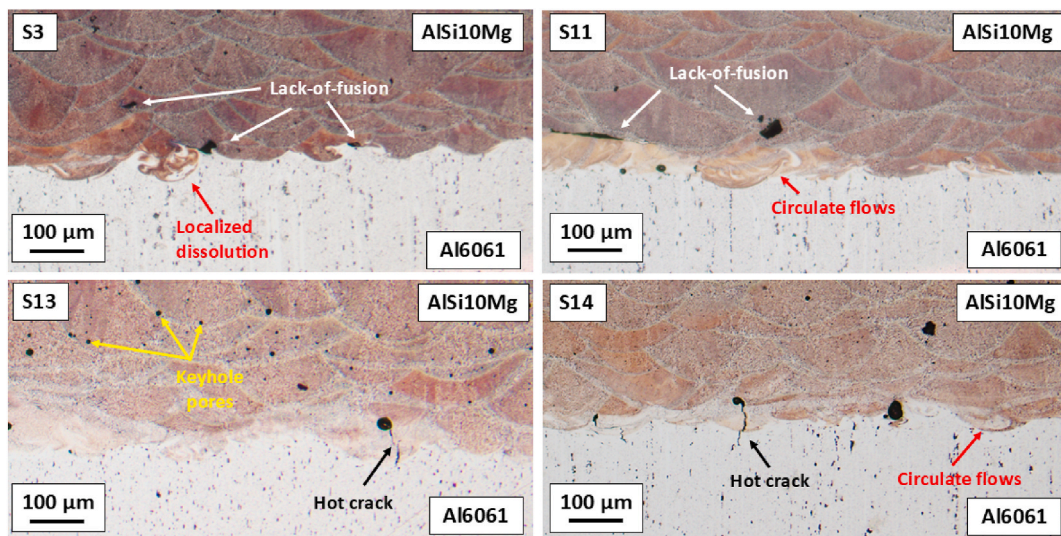


Fig. 5. Optical micrographs of etched samples 3,11,13 and 14; revealing melt pool and defect morphology in the fusion zone. All samples showed a characteristic serrated interface morphology, with circulate flows evident at the interface.

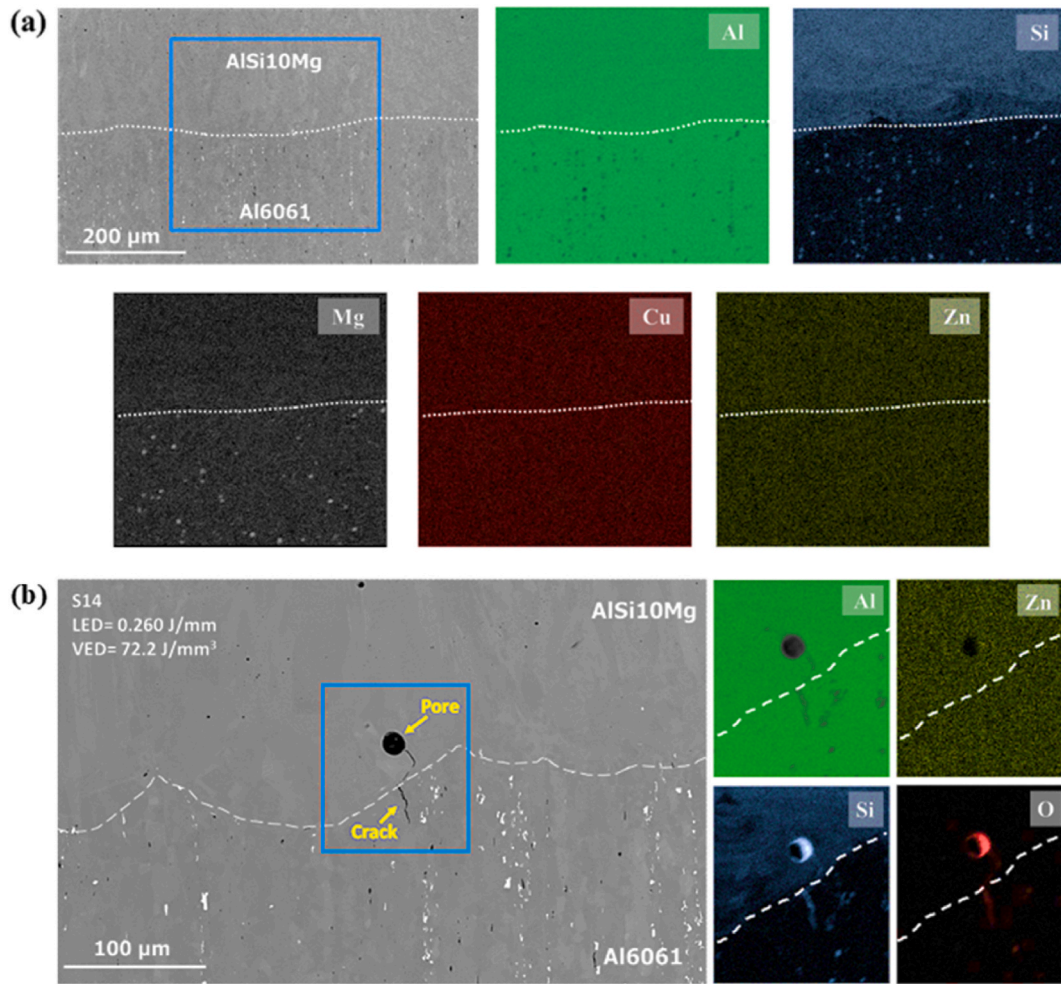


Fig. 6. (a) SEM image with the corresponding EDS maps of the Al6061/AlSi10Mg multi-material (sample 14) (b) SEM image and EDS map of a crack at the interface. The crack is emanating from a spherical pore containing SiO₂. (the white dashed line represents the interface line, and the blue box represents the region of interest). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

streak defects [30]. Furthermore, coarse and non-uniformly distributed Mg-Si particles decrease extrusion speed and can result in inadequate surface quality [31].

Table 3 shows the chemical composition, highlighting a notably higher silicon concentration on the AlSi10Mg side than on the Al6061 side. This variation reflects the alloy design, with AlSi10Mg containing approximately 9.5% silicon, which contributes to high tensile strength and hardness, while Al6061 has a lower silicon content to maintain ductility [32]. The chemical composition of Al6061/AlSi10Mg multi-material is collected from a larger area, including the interface region, and is indicated by the blue color in the SEM image. The high aluminum content in the multi-material can be attributed to the high aluminum composition of the Al6061 alloy, which typically contains around 97.60% aluminum. During the fusion process, aluminum from the Al6061 alloy can diffuse into the interface region, leading to a higher aluminum concentration.

Fig. 6(b) shows a crack that originated and propagated from a pore at the interface of the multi-material Al6061/AlSi10Mg (sample 14). EDS

analysis of the area marked by the blue box in the SEM image revealed a 0.9 wt% oxygen and 3.5 wt% silicon composition. The presence of silicon oxide in the pore likely contributed to crack formation, as the oxide phase can act as a stress concentrator, weakening the surrounding material and promoting crack propagation [33,34]. A repetitive trend of this type of crack was observed in samples 13–15. The formation of oxides could be attributed to the failure of liquid metal to fully wet the surface, which leads to poor powder consolidation and the development of oxide-driven pores from metal vapors, serving as crack initiation points [35]. Adding a minor amount of Mg and Si to the alloy could significantly reduce the powder's oxygen content, as suggested by Ghasemi et al. [36].

A compact dissolution zone with some degree of macro-segregation can be observed at the interface, as illustrated in Fig. 7, which suggests that the delivered laser energy was not sufficient to melt the Al6061 substrate for most parameter sets, despite the remelting step. This can be attributed to the high conductivity and reflectivity of Aluminum alloys, which necessitates a higher energy density to be melted [37]. Macro-segregation occurs when solute elements are rejected ahead of the solidification front but lack sufficient time for redistribution before complete solidification, a phenomenon exacerbated by the ineffective Marangoni convection at reduced heat inputs. Conversely, solidification crack formation at the interface of samples 13–15 suggests excessive melting of the Al6061 substrate. As commonly observed in the literature, AlSi10Mg solidified in an α -Al cellular

Table 3
The compositions (wt.%) of the AlSi10Mg, Al6061, and multi-material parts.

Sample	Al	Si	Mg	Cu	Zn
AlSi10Mg	90.40	9.50	0.10	–	–
Al6061	97.60	1.19	0.89	0.20	0.10
Al6061/AlSi10Mg	94.90	3.89	0.59	0.30	0.30

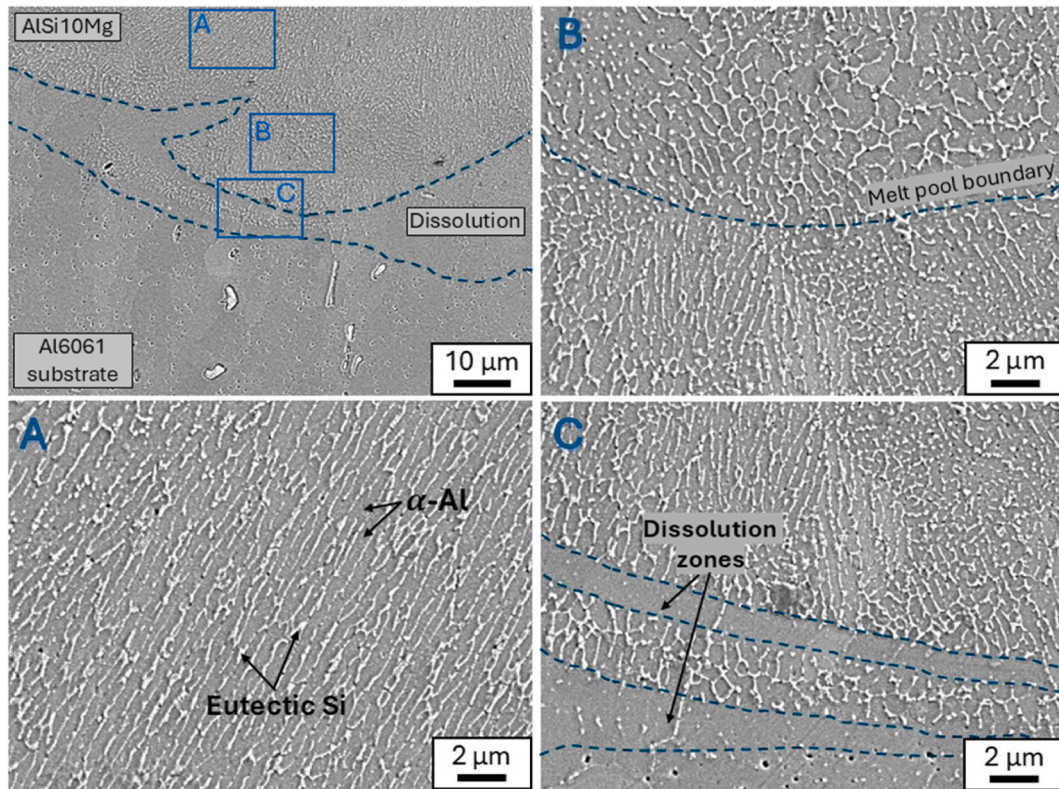


Fig. 7. High magnification images of the Al6061/AlSi10Mg fusion zone (sample 10), showing a compact dissolution zone between the two materials. A eutectic Si network exists at the boundaries of the cellular structure, which forms larger equiaxed cells at the bottom of melt pools and transitions to finer elongated cells at the center.

structure of varying cell size and texture, due to the changing heat gradient during the build process, with a dense Si network at the cell boundaries. At the bottom of the melt pool, α -Al forms a coarse equiaxed structure, which transitions to finer elongated cells at the center of the melt pool. Conversely, the dissolution zone shows a negligible presence of eutectic Si, due to the elevated temperatures which promoted Si solubility in the Al matrix [38].

Fig. 8 illustrates the EBSD inverse pole figure (IPF) map for the Al6061/AlSi10Mg multi-material part. The microstructure of AlSi10Mg is primarily composed of columnar grains, which form along the direction of the highest thermal gradient within the melt pool [39], and a minor presence of smaller equiaxed grains near the bottom of the melt pools. This pattern results in a stronger texture in the $\langle 001 \rangle$ direction and more anisotropic properties, consistent with observations reported for the microstructure of LPBF AlSi10Mg [3]. Extruded Al6061-T6 has shown a similar pattern of shorter elongated grains aligned with the extrusion direction, creating a strong texture and anisotropic mechanical properties in that direction. The T6 heat treatment further refines the microstructure by promoting the uniform distribution of precipitates, which enhances both ductility and strength [40].

Epitaxial growth was observed at both the melt pool boundary and the interface between LPBF AlSi10Mg and the Al6061 substrate, as shown in Fig. 8. This could be attributed to the remelting and material mixing [10], along with additional factors, such as temperature, grain orientation, and lattice mismatch, which could influence this growth [41]. In particular, the similarity in material composition and thermal gradients along the build direction promotes a smooth microstructural transition, reducing microscopic defects and enhancing interfacial integrity. Moreover, a notable microstructural transition occurs at the melt pool boundary, characterized by the transformation from columnar grains aligned along the build direction on the AlSi10Mg side to fine equiaxed grains with random crystallographic orientations. This

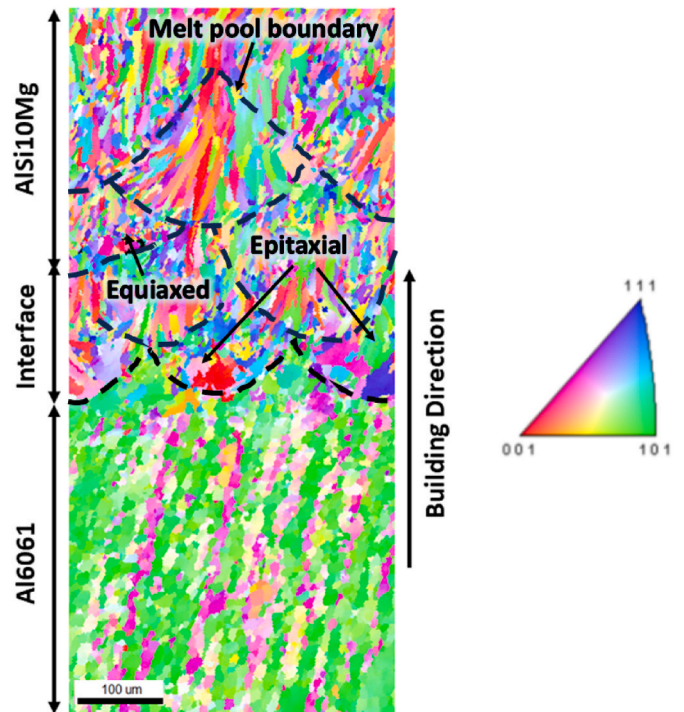


Fig. 8. EBSD IPF map for the interface of the multi-material part (the black dashed line represents the melt pool boundary).

morphological evolution is fundamentally governed by the interplay between thermal gradient (G) and solidification rate (R), which dynamically influence grain nucleation, growth, and final microstructural configuration during the LPBF process [42].

3.2. Microhardness

The microhardness measurements of the multi-material Al6061/AlSi10Mg part, as illustrated in Table 4, were conducted in three different regions: the AlSi10Mg, Al6061, and interface. The AlSi10Mg region exhibited the highest average microhardness of 109.56 ± 0.5 HV, while the Al6061 region had the lowest, at 98.20 ± 1.26 HV. The increased microhardness in the LPBF region can be attributed to the finer microstructure and sub-cell structures of AlSi10Mg [43]. The interface region, shown in Fig. 9, displays the contrasting deformation behaviors of the two alloys in the vicinity of the indentation, where the relative ‘softness’ of Al6061 resulted in a visible deformation field as opposed to AlSi10Mg. This resulted in intermediate microhardness values between the two alloys, averaging 99.35 ± 1.03 HV, slightly higher than the Al6061 region. This gradual increase in hardness at the interface results from the partial melting of Al6061, its mixing with molten AlSi10Mg, and the solidification of an alloy enhanced with alloying elements, forming a hardened layer through solid solution strengthening [44]. The gradient cellular microstructure observed in LPBF parts (see Fig. 7), characterized by finer structures within melt pools and coarser structures at the boundaries, leads to variations in hardness across the interface. This variation in microhardness is influenced by the solubility of Si in the Al matrix, which is affected by energy density during the LPBF process; higher energy densities can enhance Si solubility and create localized hardness differences [38]. Optimizing process parameters to achieve nearly defect-free components significantly contributes to the overall hardness profile by minimizing porosity-induced softening effects.

3.3. Shear test

Preliminary tensile tests conducted on the Al6061/AlSi10Mg hybrid components showed consistent failure within the Al6061 region away from the interface, at an average ultimate tensile strength of 329.7 MPa and an elongation of 2.41 %. This occurs because tensile loading places the substrate, interface, and LPBF parts in an isostress state, where tensile failure occurs in the weakest region (Al6061). This observation aligns with the findings of several studies [10,12,15,16] on similar hybrid components, reporting failure in the weaker base material rather than at the interface. Therefore, in this study, shear testing was chosen to focus the load on the fusion area and quantitatively correlate the bonding strength to the chosen processing parameters.

Table 5 lists the shear strengths of the L16 orthogonal set of multi-material samples. Shear tests performed on neat Al6061 and AlSi10Mg specimens showed that Al6061 achieved the highest average shear strength, reaching 222.37 ± 1.69 MPa, while AlSi10Mg exhibited an average value of 183.69 ± 4.43 MPa. Note that the shear strength obtained for Al6061-T6 agrees well with literature values of 207–222 MPa [23]. Among the multi-material samples, sample 14 ($P = 325$ W, $v = 1250$ mm/s, $h = 0.12$ mm) showed the highest average shear strength of 178.54 ± 5.49 MPa, slightly less than pure AlSi10Mg.

Fig. 10(a) compares the shear stress vs. strain curve of the multi-

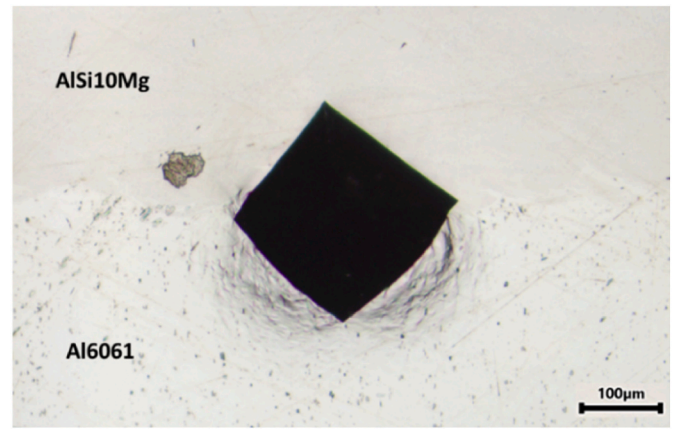


Fig. 9. Optical micrograph of the hardness indentation at the interface of the multi-material part.

Table 5
Interfacial porosity and shear strength results for the L16 multi-material samples.

Sample no.	Power [W]	Scanning Speed [mm/s]	Hatching Distance [mm]	Interface Porosity [%]	Shear Strength [MPa]
1	250	1000	0.06	0.58	166.8 ± 4.03
2	250	1250	0.09	0.3	170.7 ± 6.99
3	250	1500	0.12	3.35	–
4	250	1750	0.15	4.79	–
5	275	1000	0.09	0.23	173.8 ± 6.32
6	275	1250	0.06	0.59	165.7 ± 5.25
7	275	1500	0.15	0.44	168.2 ± 3.42
8	275	1750	0.12	0.35	169.6 ± 2.34
9	300	1000	0.12	0.07	173.2 ± 4.29
10	300	1250	0.15	0.01	171.5 ± 4.25
11	300	1500	0.06	0.23	170.3 ± 4.50
12	300	1750	0.09	0.16	176.9 ± 3.58
13	325	1000	0.15	0.12	175.4 ± 2.24
14	325	1250	0.12	0.07	178.5 ± 5.49
15	325	1500	0.09	0.21	172.6 ± 5.70
16	325	1750	0.06	0.22	168.0 ± 1.66

material sample 14 to those obtained for the neat Al6061 and AlSi10Mg samples. Note that the Al6061 stress-strain curve was truncated due to high ductility. Al6061 showed the greatest elongation, highlighting its superior ductility, whereas AlSi10Mg had the lowest elongation, reflecting its more brittle nature. Meanwhile, the multi-material sample showed intermediate elongation, suggesting a balance between the ductility of Al6061 and the brittleness of AlSi10Mg. An intermediate performance of the multi-material parts would also be expected in terms of shear strength. However, the multi-material Al6061/AlSi10Mg parts consistently exhibited a shear strength lower than that of AlSi10Mg. This ‘under-performance’ in terms of shear strength can be tightly correlated with interfacial LOF porosity, which limits the mechanical performance of the multi-material structure [15].

Table 4

Hardness values at three different points in the multi-material part for the Al6061, interface, and AlSi10Mg regions.

Region	Point 1 [HV]	Point 2 [HV]	Point 3 [HV]
Al6061	97.08	97.80	99.73
Interface	99.98	98.06	100.02
AlSi10Mg	110.03	109.70	108.97

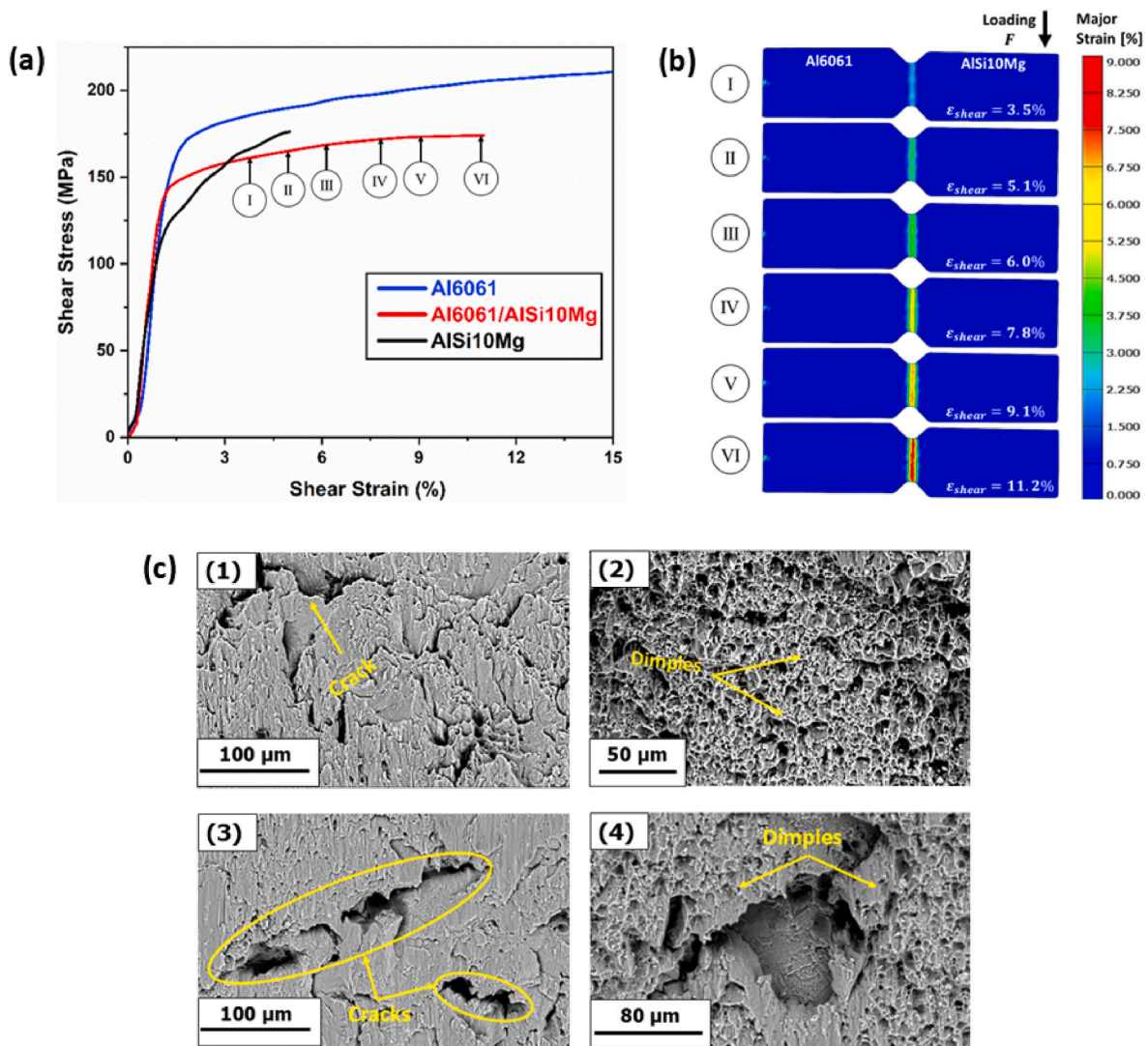


Fig. 10. (a) Iosipescu Shear Stress-Strain curves for the Al6061, Al6061/AlSi10Mg (sample 14), and AlSi10Mg parts. (b) Time stamps show the progression of major strain during the shear test. (c) SEM images showing the fracture morphologies of the tested Al6061/AlSi10Mg multi-material part (sample 14) where (1) and (2) show the Al6061 side, and (3) and (4) show the AlSi10Mg side.

Indeed, reducing or controlling porosity at the interface could improve the shear strength of such multi-material components [45].

Fig. 10(b) displays major strain progression maps in the multi-material Al6061/AlSi10Mg sample with increasing load, generated through GOM software. These maps illustrate that strain becomes increasingly concentrated at the interface between the two alloys as the test progresses, demonstrating that the interface experiences higher shear deformations under the applied load than other regions. Despite the observed strain localization, the shear strain remains relatively uniform across the length of the interface. The shear bandwidth increases with progressive loading throughout the test. The highest strain remains localized at the center, surrounded by bands of gradually decreasing strain.

The shear properties of the Al6061/AlSi10Mg hybrid part are intricately linked to its microstructure, which is considerably influenced by the LPBF process parameters. The observed epitaxial growth at the interface and melt pool boundaries in Fig. 8 plays a crucial role in determining interfacial strength. The transition from columnar grains in the AlSi10Mg region to fine equiaxed grains at the interface promotes uniform stress distribution, enhancing mechanical properties [41]. However, the narrow fusion zone ($<20\text{ }\mu\text{m}$), observed in samples built at low energy density, introduces a sharp variation in composition and

microstructure which promotes localized stress concentration in this region during mechanical loading. Similarly, macro-segregation of ‘islands’ at the interface may result in a similar delamination behavior [46], decreasing the mechanical strength of the part. Alternatively, the wider fusion zone in samples 13–15, provided a more gradual variation in composition which resulted in superior bonding ($>172\text{ MPa}$), despite the presence of solidification cracks within the fusion zone. The orthogonal alignment of the cracks relative to the shear loading direction may be the primary cause behind their limited impact on shear strength.

Fig. 10(c) presents SEM images of the fracture surfaces of the tested multi-material Al6061/AlSi10Mg samples. While both sides of the interface exhibit cracks and dimples, some differences can be observed in the fracture characteristics. The Al6061 side generally shows a higher density of dimples, which may suggest a greater extent of plastic deformation prior to failure, consistent with ductile fracture features such as microvoid coalescence [47]. On the other hand, the AlSi10Mg side presents fewer dimples and a relatively higher presence of cracks, which could indicate a tendency towards brittle fracture, potentially influenced by the presence of pores and silicon oxide particles acting as stress concentrators [48–51]. However, it can be noticed that both sides exhibit a mix of fracture features, and distinguishing a significantly

Table 6

Comparison between the joint strength and elongation of dissimilar aluminum joints, evaluated through various mechanical testing methods.

Material A (manufacturing method)	Material B (manufacturing method)	Joining Process	Testing Method	Ultimate Strength [MPa]	Elongation [%]	Ref.
Al6061-T6 (Extruded)	AlSi10Mg	LPBF	Iosipescu Shear	178.54 ± 5.49	11.2	Present Study
AA1050	AlSi10Mg (LPBF)	Laser Welding	Lap Shear	185	–	[52]
AA6082 (Rolled)	AlSi10Mg (LPBF)	Laser Welding	Lap Shear	204 ± 9.8	–	[53]
Al6061-T6 (Hot Rolled)	AlSi10Mg (LPBF)	Friction Stir Welding	Tensile	206	8	[54]
AA6061 (Wrought)	AlSi10Mg	LPBF	Tensile	192.8 ± 3.4	13.6 ± 0.5	[12]
AA2618 (Cast)	AlSi10Mg	LPBF	Tensile	187 ± 5	3	[44]

different fracture mode requires further detailed analysis.

Table 6 compares the joint strength and elongation of hybrid aluminum parts fabricated using various techniques and evaluated through different testing methods. While current research on multi-material joints by LPBF primarily utilizes tensile testing, the present study is the first to employ the Iosipescu shear test in this field. This approach addresses the gap of shear testing in hybrid aluminum LPBF parts, despite its frequent use in traditional welding. In the present study, the Iosipescu shear test for the Al6061/AlSi10Mg joint resulted in an ultimate shear strength of 178.54 ± 5.49 MPa, which is comparable to the shear strength of laser-welded AA1050/AlSi10Mg [52] (185 MPa) but slightly lower than the shear strength of the AA6082/AlSi10Mg joint [53] (204 ± 9.8 MPa). This indicates that the Iosipescu shear test is a reliable technique for evaluating interfacial shear behavior, which is crucial for multi-material applications where shear loading is a key consideration.

In this study, a correlation between interfacial porosity (in %) and shear strength was observed in the samples studied, as shown in Fig. 11. The samples show a general trend of decreasing shear strength with increasing interfacial porosity. Samples 13 and 14 had two of the highest shear strength values despite the presence of small-diameter keyhole pores and vertical microcracks, indicating the low influence of these defects on the multi-material parts' shear strength compared to larger LOF voids.

3.4. Analysis of the effect of process parameters

Fig. 12 shows 3D plots of the dependency of shear strength and

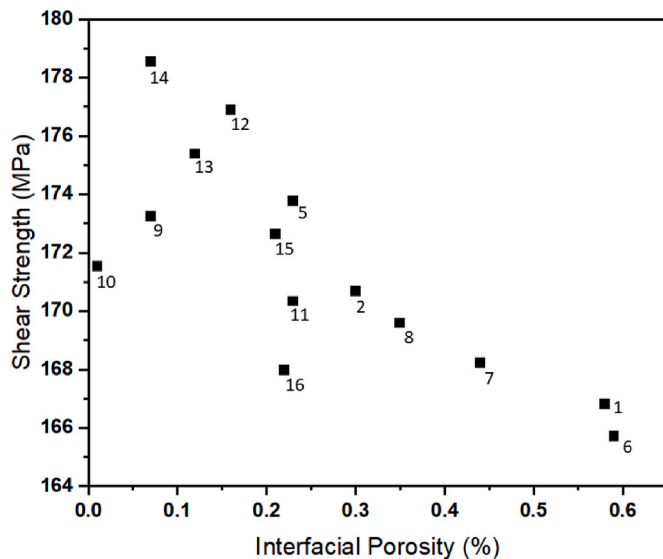


Fig. 11. Plot of shear strength against interfacial porosity for the tested samples, where a general decreasing trend in shear strength is observed with increasing interfacial porosity.

interfacial porosity on the processing parameters. A 3rd degree polynomial surface was used to fit the 14 data points with a Root Mean Squared Error (RMSE) of 2.443, 1.300, and 1.714 for the shear strength plots, respectively, and an RMSE of 0.0764, 0.0458, 0.0601 for the porosity plots, respectively. From the plots, it is evident that a combination of minimum scanning speed, maximum laser power, and a hatching space between 0.09 and 0.12 mm resulted in enhanced bonding between the substrate and additive part, improving the shear strength of the interface. In the case of interfacial porosity, high laser power is imperative for a high-density interface. Higher porosity was also obtained at the extremities of the tested range of hatch spacing and scanning speeds. The line of minimum porosity followed a diagonal path between hatch spacing and scanning speeds.

Taguchi analysis was also performed on the L16 set to identify the isolated effect of each process parameter on the porosity and shear strength of the multi-material parts, respectively. The main effects plots are illustrated in Fig. 13(a) for porosity and Fig. 13(b) for shear strength. Fig. 13(a) shows a general decrease in porosity with increasing power. The scanning speed has a minimal effect on porosity at lower values of the scanning speed, 1000 and 1250 mm/s, reaching a minimum threshold of 0.3 %. Increasing the scanning speed beyond 1250 mm/s increases porosity drastically. A hatching space of 90 μ m is suggested to ensure minimum porosity, while larger hatching distances contribute to a higher porosity percentage. According to the Taguchi results, minimum interfacial porosity can be obtained at 300 W, 1250 mm/s, and 90 μ m.

Shear strength shows a similar response to power, where increasing power leads to higher shear strength. This could be attributed to the deeper melt pools that form at high laser power, inducing stronger overlap and enhanced fusion between subsequent layers [3]. Scanning speed decreases to a minimum at 1500 mm/s before increasing again at 1750 mm/s. In contrast, the highest shear strength may be achieved at a hatching space of 90 and 120 μ m. Considering these results, the maximum shear strength is expected at 325W, 1000 mm/s, and 120 μ m.

One must note the discrepancy between the optimized parameters for the lowest porosity and maximum shear strength, where achieving one does not necessitate achieving the other. Nonetheless, observing the Taguchi analysis in Fig. 13, using the parameters optimized for the lowest porosity, is also expected to achieve relatively high shear strength. It is also worth noting that the interactions between variables may sometimes be misinterpreted by Taguchi designs, leading to what is known as 'confounding variables' where one variable may mask the effect of others.

Table 7 depicts the response table for means of porosity and shear strength, where the rank shows the relative sensitivity of the means to each process parameter. The results indicate that laser power primarily influenced interfacial porosity, followed by scanning speed and hatching space. The findings agree with the microstructural observations in section 3.1. In contrast, the shear strength is influenced mainly by the hatching space, followed by power, and the least influenced by scanning speed.

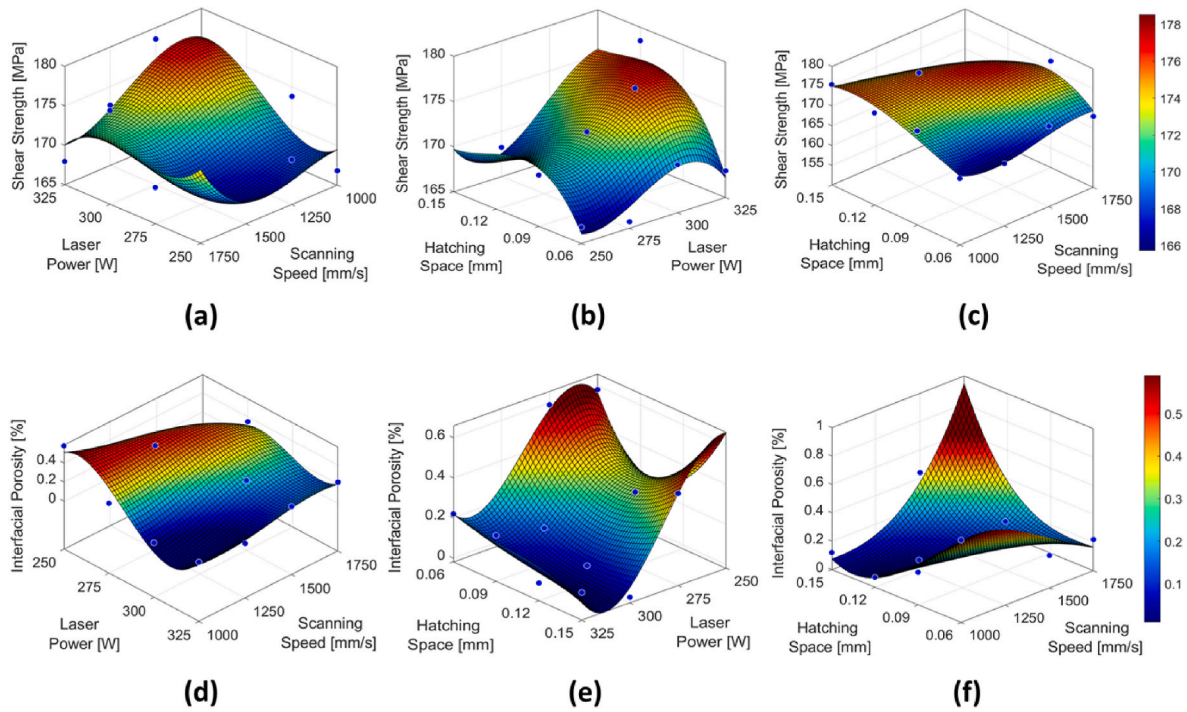


Fig. 12. 3D surface plots highlighting the relationships between the three process parameters and each response variable: (a–c) shear strength and (d–f) interfacial porosity. A polynomial function of the 3rd degree was used to fit the points to a 3D surface.

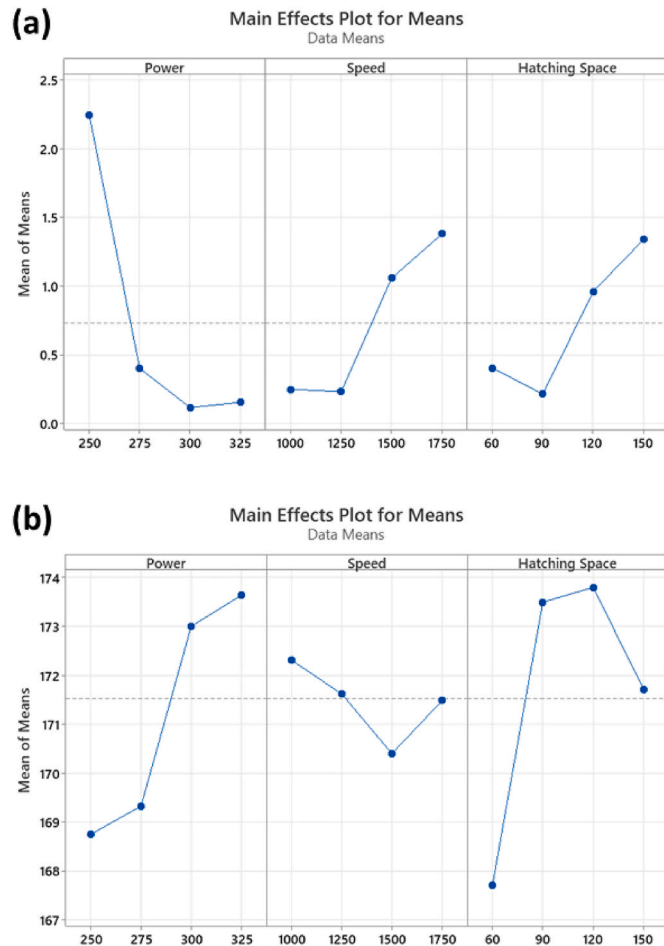


Fig. 13. Taguchi analysis plots for (a) porosity and (b) shear strength.

Table 7

Taguchi response table for means.

Level	Porosity [%]			Shear Strength [MPa]		
	Power	Speed	Hatching Space	Power	Speed	Hatching Space
1	2.2450	0.2500	0.4050	168.7	172.3	167.7
2	0.4025	0.2325	0.2150	169.3	171.6	173.5
3	0.1175	1.0575	0.9600	173.0	170.4	173.8
4	0.1550	1.3800	1.3400	173.6	171.5	171.7
Delta	2.1275	1.1475	1.1250	4.9	1.9	6.1
Rank	1	2	3	2	3	1

4. Conclusions

This study investigated the microstructure and interfacial mechanical properties of the multi-material Al6061/AlSi10Mg parts fabricated by LPBF, with varying laser parameters specifically applied to the first ten layers of the interface (0.3 mm in thickness). The main findings are as follows:

- The insufficiency of LED and VED as independent indicators of defect morphology is observed. This is evidenced by a contrary-to-expectation transition from lack of fusion (LOF) defects to keyhole pores and hot cracks with the decrease of VED, observed at a constant scanning speed of 1000 mm/s. Similarly, it is evidenced by the simultaneous formation of both LOF defects and keyhole pores in samples with identical LED but different powers.
- Defect type and morphology were primarily influenced by laser power, followed by LED/scanning speed, then hatching space, as evident from microstructural characterization and Taguchi analysis.
- In contrast, hatch spacing was a significant factor in determining the shear strength of the bimetallic parts, with optimum results achieved at $h = 90 - 120 \mu\text{m}$.
- The fracture morphology under shear loading shows a ductile failure on the Al6061 side with significant plastic deformation, while the

AlSi10Mg side demonstrates more brittle fracture due to pores and inclusions acting as stress concentrators.

- The highest average shear strength of the Al6061/AlSi10Mg multi-material parts was 178.54 ± 5.49 MPa. A negative correlation between shear strength and interfacial porosity was observed. Additionally, the presence of longitudinal solidification cracks ($<100 \mu\text{m}$) showed a negligible effect on shear strength, due to their orthogonal orientation to the shear loading direction.
- The AlSi10Mg region achieved the highest microhardness (109.56 ± 0.5 HV), while the Al6061 region exhibited the lowest (98.20 ± 1.26 HV). The interface showed intermediate hardness (99.35 ± 1.03 HV) due to partial melting and mixing of the two alloys.

Future research should be aimed at fine-tuning laser parameters to minimize the LOF and keyhole porosity in multi-material Al6061/AlSi10Mg structures. Moreover, employing a variety of approaches, such as pre-processing (e.g., surface treatments), remelting, and post-processing (e.g., heat treatments), might provide valuable insights into further enhancing these multi-material components' interfacial integrity and mechanical properties.

CRedit authorship contribution statement

Linda Ismail: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Omer F. Mohamed:** Writing – original draft, Visualization, Investigation, Data curation. **Tareq Farrah:** Writing – review & editing, Resources, Methodology. **Pradeep George:** Writing – review & editing, Validation, Investigation. **Andreas Schiffer:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon request.

References

- [1] K.V. Yang, P. Rometsch, T. Jarvis, J. Rao, S. Cao, C. Davies, X. Wu, Porosity formation mechanisms and fatigue response in Al-Si-Mg alloys made by selective laser melting, *Mater. Sci. Eng., A* 712 (2018) 166–174, <https://doi.org/10.1016/j.msea.2017.11.078>.
- [2] D. Buchbinder, W. Meiners, K. Wissenbach, R. Poprawe, Selective laser melting of aluminum die-cast alloy—correlations between process parameters, solidification conditions, and resulting mechanical properties, *J. Laser Appl.* 27 (2015) 29205, <https://doi.org/10.2351/1.4906389/313998>.
- [3] N.T. Aboulkhair, M. Simonelli, L. Parry, I. Ashcroft, C. Tuck, R. Hague, 3D printing of Aluminium alloys: additive Manufacturing of Aluminium alloys using selective laser melting, *Prog. Mater. Sci.* 106 (2019) 100578, <https://doi.org/10.1016/j.pmatsci.2019.100578>.
- [4] A.H. Maamoun, Y.F. Xue, M.A. Elbestawi, S.C. Veldhuis, Effect of selective laser melting process parameters on the quality of Al alloy parts: powder characterization, density, surface roughness, and dimensional accuracy, *Materials* 11 (2018) 2343, <https://doi.org/10.3390/MA1122343>, 11 (2018) 2343.
- [5] X. Gu, J. Zhang, X. Fan, N. Dai, Y. Xiao, L.C. Zhang, Abnormal corrosion behavior of selective laser melted AlSi10Mg alloy induced by heat treatment at 300 °C, *J. Alloys Compd.* 803 (2019) 314–324, <https://doi.org/10.1016/j.jallcom.2019.06.274>.
- [6] M. Vaezi, S. Chianrabutra, B. Mellor, S. Yang, Multiple material additive manufacturing – Part 1: a review, *Virtual Phys. Prototyp.* 8 (2013) 19–50, <https://doi.org/10.1080/17452759.2013.778175>.
- [7] A. Bandyopadhyay, B. Heer, Additive manufacturing of multi-material structures, *Mater. Sci. Eng. R Rep.* 129 (2018) 1–16, <https://doi.org/10.1016/j.mser.2018.04.001>.
- [8] D. Gu, X. Shi, R. Poprawe, D.L. Bourell, R. Setchi, J. Zhu, Material-structure-performance integrated laser-metal additive manufacturing, *Science* (2021) 372, https://doi.org/10.1126/SCIENCE.ABG1487/ASSET/C9C9DF6C-17D9-4FC4-9AD2-001ACE3899FC/ASSETS/GRAPHIC/372_ABG1487_F5.JPEG, 1979.
- [9] C. Wei, L. Li, Recent progress and scientific challenges in multi-material additive manufacturing via laser-based powder bed fusion, *Virtual Phys. Prototyp.* 16 (2021) 347–371, <https://doi.org/10.1080/17452759.2021.1928520>.
- [10] A. Hadadzadeh, B.S. Amirkhiz, S. Shakerin, J. Kelly, J. Li, M. Mohammadi, Microstructural investigation and mechanical behavior of a two-material component fabricated through selective laser melting of AlSi10Mg on an Al-Cu-Ni-Fe-Mg cast alloy substrate, *Addit. Manuf.* 31 (2020), <https://doi.org/10.1016/j.addma.2019.100937>.
- [11] P. Fathi, M. Rafieazad, E. Mohseni-Sohi, M. Sanjari, H. Pirgazi, B. Shalchi Amirkhiz, M. Ghoncheh, A. Nasiri, M. Mohammadi, Corrosion performance of additively manufactured bimetallic aluminum alloys, *Electrochim. Acta* 389 (2021), <https://doi.org/10.1016/j.electacta.2021.138689>.
- [12] C. Zhao, Y. Bai, H. Wang, Feasibility and reliability of laser powder bed fused AlSi10Mg/wrought AA6061 hybrid aluminium alloy component, *Int. J. Precision Eng. Manuf. Green Technol.* (2022), <https://doi.org/10.1007/s40684-022-00456-6>.
- [13] X. Tian, Z. Zhao, H. Wang, X. Liu, X. Song, Progresses on the additive manufacturing of functionally graded metallic materials, *J. Alloys Compd.* 960 (2023) 170687, <https://doi.org/10.1016/j.jallcom.2023.170687>.
- [14] S.L. Sing, S. Huang, G.D. Goh, G.L. Goh, C.F. Tey, J.H.K. Tan, W.Y. Yeong, Emerging metallic systems for additive manufacturing: in-situ alloying and multi-metal processing in laser powder bed fusion, *Prog. Mater. Sci.* 119 (2021) 100795, <https://doi.org/10.1016/j.pmatsci.2021.100795>.
- [15] D. Bhaduri, P. Penchev, K. Essa, S. Dimov, L.N. Carter, C.I. Pruncu, D. Pullini, Evaluation of surface/interface quality, microstructure and mechanical properties of hybrid additive-subtractive aluminium parts, *CIRP Annals* 68 (2019) 237–240, <https://doi.org/10.1016/j.cirp.2019.04.116>.
- [16] H. Wang, X. Lin, N. Kang, Z. Qin, S. Shi, J. Li, W. Huang, Interfacial characteristics and mechanical behavior of hybrid manufactured AlSi10Mg–Al6061 bimetal via selective laser melting and forging, *Acta Metall. Sin.* 35 (2022) 375–388, <https://doi.org/10.1007/S40195-021-01349-4/TABLES/5>.
- [17] G. Zhu, L. Wang, B. Wang, B. Li, J. Zhao, B. Ding, R. Cui, B. Jiang, C. Zhao, B. Su, L. Luo, R. Chen, Y. Su, J. Guo, Multi-materials additive manufacturing of Ti6Al4V/Cu/316L by electron beam freeform fabrication, *J. Mater. Res. Technol.* 26 (2023) 8388–8405, <https://doi.org/10.1016/j.jmrt.2023.09.194>.
- [18] F. Bartolomeu, M.M. Costa, N. Alves, G. Miranda, F.S. Silva, Additive manufacturing of NiTi-Ti6Al4V multi-material cellular structures targeting orthopedic implants, *Opt Laser. Eng.* 134 (2020) 106208, <https://doi.org/10.1016/j.optlaseng.2020.106208>.
- [19] I.H. ZainElabdeen, L. Ismail, O.F. Mohamed, K.A. Khan, A. Schiffer, Recent advancements in hybrid additive manufacturing of similar and dissimilar metals via laser powder bed fusion, *Mater. Sci. Eng., A* 909 (2024) 146833, <https://doi.org/10.1016/j.msea.2024.146833>.
- [20] U. Khashaba, T. Sebaey, F. Mahmoud, A. Selmy, R. Hamouda, Experimental and numerical scaling of pinned-joints composite laminates: effects of stacking sequences, *J. Compos. Mater.* 47 (2013) 3353–3366, <https://doi.org/10.1177/0021998312464891>.
- [21] Standard test method for shear properties of composite materials by the V-notched beam method, Pennsylvania. <https://doi.org/10.1520/D5379D5379M-19E01>, 1999.
- [22] D.E. Walrath, D.F. Adams, *Verification and Application of the Iosipescu Shear Test Method*, 1984. Virginia.
- [23] Ö. Ünal, D.J. Barnard, I.E. Anderson, A shear test method to measure shear strength of metallic materials and solder joints using small specimens, *Acta Metall.* 40 (1998).
- [24] A. Sola, A. Nouri, Microstructural porosity in additive manufacturing: the formation and detection of pores in metal parts fabricated by powder bed fusion, *J Adv Manuf Process* 1 (2019), <https://doi.org/10.1002/amp.2.10021>.
- [25] Z. Gan, O.L. Kafka, N. Parab, C. Zhao, L. Fang, O. Heinonen, T. Sun, W.K. Liu, Universal scaling laws of keyhole stability and porosity in 3D printing of metals, *Nat. Commun.* 12 (2021) 2379, <https://doi.org/10.1038/s41467-021-22704-0>.
- [26] H. Pan, A. Witvrouw, F. Wolf-Monheim, R. Soudchek, S. Yang, Effects of substrate surface treatments on hybrid manufacturing of AlSi7Mg using die casting and selective laser melting, *J. Mater. Sci. Technol.* (2023), <https://doi.org/10.1016/j.jmst.2023.02.009>.
- [27] Z. Kuai, Z. Li, B. Liu, Y. Chen, H. Li, P. Bai, Microstructure and mechanical properties of CuCrZr/316L hybrid components manufactured using selective laser melting, *J. Alloys Compd.* 955 (2023) 170103, <https://doi.org/10.1016/j.jallcom.2023.170103>.
- [28] I. Bitharas, N. Parab, C. Zhao, T. Sun, A.D. Rollett, A.J. Moore, The interplay between vapour, liquid, and solid phases in laser powder bed fusion, *Nat. Commun.* 13 (2022), <https://doi.org/10.1038/s41467-022-30667-z>.
- [29] K. Singh, V.K. Sharma, T. Singh, A. Kumar Singh, R. Singh, Correlating surface roughness with cutting temperature in face milling of Al-6061 using CRITIC

- approach, *Mater. Today Proc.* 69 (2022) 333–338, <https://doi.org/10.1016/j.matpr.2022.08.545>.
- [30] Y. Birol, The effect of homogenization practice on the microstructure of AA6063 billets, *J. Mater. Process. Technol.* 148 (2004) 250–258, <https://doi.org/10.1016/J.JMATPROTEC.2004.01.056>.
- [31] H. Zhu, M.J. Couper, A.K. Dahle, Effect of process variables on Mg-Si particles and extrudability of 6xxx series aluminum extrusions, *J. Occup. Med.* 63 (2011) 66–71, <https://doi.org/10.1007/S11837-011-0183-2/METRICS>.
- [32] K.G. Prashanth, S. Scudino, H.J. Klauss, K.B. Surreddi, L. Löber, Z. Wang, A. K. Chaubey, U. Kühn, J. Eckert, Microstructure and mechanical properties of Al–12Si produced by selective laser melting: effect of heat treatment, *Mater. Sci. Eng., A* 590 (2014) 153–160, <https://doi.org/10.1016/J.MSEA.2013.10.023>.
- [33] Y. Zhang, A. Majeed, M. Muzamil, J. Lv, T. Peng, V. Patel, Investigation for macro mechanical behavior explicitly for thin-walled parts of AlSi10Mg alloy using selective laser melting technique, *J. Manuf. Process.* 66 (2021) 269–280, <https://doi.org/10.1016/J.JMAPRO.2021.04.022>.
- [34] E. Louvis, P. Fox, C.J. Sutcliffe, Selective laser melting of aluminium components, *J. Mater. Process. Technol.* 211 (2011) 275–284, <https://doi.org/10.1016/J.JMATPROTEC.2010.09.019>.
- [35] M. Tang, P.C. Pistorius, Oxides, porosity and fatigue performance of AlSi10Mg parts produced by selective laser melting, *Int. J. Fatig.* 94 (2017) 192–201, <https://doi.org/10.1016/J.IJFATIGUE.2016.06.002>.
- [36] A. Ghasemi, E. Fereiduni, M. Balbaa, S.D. Jadhav, M. Elbestawi, S. Habibi, Influence of alloying elements on laser powder bed fusion processability of aluminum: a new insight into the oxidation tendency, *Addit. Manuf.* 46 (2021) 102145, <https://doi.org/10.1016/J.ADDMA.2021.102145>.
- [37] N.T. Aboulkhair, M. Simonelli, L. Parry, I. Ashcroft, C. Tuck, R. Hague, 3D printing of Aluminium alloys: additive Manufacturing of Aluminium alloys using selective laser melting, *Prog. Mater. Sci.* 106 (2019), <https://doi.org/10.1016/j.pmatsci.2019.100578>.
- [38] L. Zhou, A. Mehta, E. Schulz, B. McWilliams, K. Cho, Y. Sohn, Microstructure, precipitates and hardness of selectively laser melted AlSi10Mg alloy before and after heat treatment, *Mater. Char.* 143 (2018) 5–17, <https://doi.org/10.1016/J.MATCHAR.2018.04.022>.
- [39] U. Patakhham, A. Palasay, P. Wila, R. Tongsri, MPB characteristics and Si morphologies on mechanical properties and fracture behavior of SLM AlSi10Mg, *Mater. Sci. Eng., A* 821 (2021) 141602, <https://doi.org/10.1016/J.MSEA.2021.141602>.
- [40] P. Zhang, S. Zhao, C. Zhang, Z. Chen, J. Zhang, L. Fei, P. Dong, Microstructure and mechanical properties of probeless friction stir extrusion joined joints of 6061-T6 aluminum alloy to Q235 steel, *Int. J. Adv. Manuf. Technol.* 119 (2022) 3029–3043, <https://doi.org/10.1007/S00170-021-08363-X/FIGURES/13>.
- [41] A. Hadadzadeh, B.S. Amirkhiz, J. Li, M. Mohammadi, Columnar to equiaxed transition during direct metal laser sintering of AlSi10Mg alloy: effect of building direction, *Addit. Manuf.* 23 (2018) 121–131, <https://doi.org/10.1016/J.ADDMA.2018.08.001>.
- [42] I. Markov, S. Stoyanov, Mechanisms of epitaxial growth, *Contemp. Phys.* 28 (1987) 267–320, <https://doi.org/10.1080/00107518708219073>.
- [43] Q. Yan, B. Song, Y. Shi, Comparative study of performance comparison of AlSi10Mg alloy prepared by selective laser melting and casting, *J. Mater. Sci. Technol.* 41 (2020) 199–208, <https://doi.org/10.1016/J.JMST.2019.08.049>.
- [44] M.H. Ghoncheh, M. Sanjari, E. Cyr, J. Kelly, H. Pirgazi, S. Shakerin, A. Hadadzadeh, B.S. Amirkhiz, L.A.I. Kestens, M. Mohammadi, On the solidification characteristics, deformation, and functionally graded interfaces in additively manufactured hybrid aluminum alloys, *Int. J. Plast.* 133 (2020) 102840, <https://doi.org/10.1016/J.IJPLAS.2020.102840>.
- [45] N.T. Aboulkhair, N.M. Everitt, I. Ashcroft, C. Tuck, Reducing porosity in AlSi10Mg parts processed by selective laser melting, *Addit. Manuf.* 1–4 (2014) 77–86, <https://doi.org/10.1016/J.ADDMA.2014.08.001>.
- [46] S. Shakerin, A. Hadadzadeh, B.S. Amirkhiz, S. Shamsdini, J. Li, M. Mohammadi, Additive manufacturing of maraging steel-H13 bimetal using laser powder bed fusion technique, *Addit. Manuf.* 29 (2019), <https://doi.org/10.1016/j.addma.2019.100797>.
- [47] M.R. Rezaei, M.R. Toroghinejad, F. Ashrafzadeh, Effects of ARB and ageing processes on mechanical properties and microstructure of 6061 aluminum alloy, *J. Mater. Process. Technol.* 211 (2011) 1184–1190, <https://doi.org/10.1016/J.JMATPROTEC.2011.01.023>.
- [48] I. Rosenthal, A. Stern, N. Frage, Strain rate sensitivity and fracture mechanism of AlSi10Mg parts produced by Selective Laser Melting, *Mater. Sci. Eng., A* 682 (2017) 509–517, <https://doi.org/10.1016/J.MSEA.2016.11.070>.
- [49] W. Li, S. Li, J. Liu, A. Zhang, Y. Zhou, Q. Wei, C. Yan, Y. Shi, Effect of heat treatment on AlSi10Mg alloy fabricated by selective laser melting: microstructure evolution, mechanical properties and fracture mechanism, *Mater. Sci. Eng., A* 663 (2016) 116–125, <https://doi.org/10.1016/J.MSEA.2016.03.088>.
- [50] B. Amir, E. Kochavi, S. Gruntman, Y. Gale, S. Samuha, O. Sadot, Experimental investigation on shear strength of laser powder bed fusion AlSi10Mg under quasi-static and dynamic loads, *Addit. Manuf.* 46 (2021), <https://doi.org/10.1016/j.addma.2021.102150>.
- [51] S. Chen, Q. Tan, W. Gao, G. Wu, J. Fan, Z. Feng, T. Huang, A.W. Godfrey, M. Zhang, X. Huang, Effect of heat treatment on the anisotropy in mechanical properties of selective laser melted AlSi10Mg, *Mater. Sci. Eng., A* 858 (2022) 144130, <https://doi.org/10.1016/J.MSEA.2022.144130>.
- [52] T. Sun, C. Liu, P. Franciosa, N. Ferguson, G. Gibbons, D. Ceglarek, E. Mogire, P. Zhang, Effect of placement configuration on the microstructure, porosity and mechanical performance of dissimilar remote laser welding of additive manufactured AlSi10Mg alloy and conventionally manufactured 1050 aluminium sheet, *J. Mater. Res. Technol.* 27 (2023) 5639–5650, <https://doi.org/10.1016/J.JMRT.2023.11.077>.
- [53] E. Liverani, A. Ascari, A. Fortunato, The role of filler wire and scanning strategy in laser welding of difficult-to-weld aluminum alloys, *Int. J. Adv. Manuf. Technol.* 128 (2023) 763–777, <https://doi.org/10.1007/S00170-023-11932-X>.
- [54] A. Thakur, V. Sharma, N. Minhas, S. Manda, V.S. Sharma, Microstructure and mechanical properties of dissimilar friction stir welded joints of laser powder bed fusion processed AlSi10Mg and conventional hot rolled 6061-T6 thin sheets, *Opt Laser. Technol.* 163 (2023) 109382, <https://doi.org/10.1016/J.OPTLASTEC.2023.109382>.